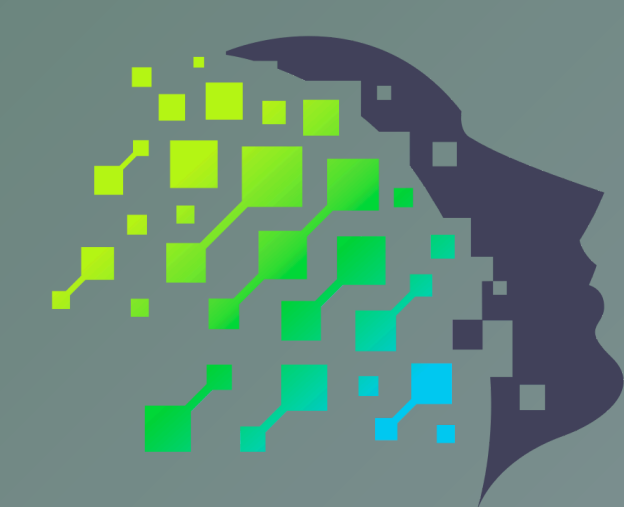


Learning Geometric Reasoning Networks for Robot Task and Motion Planning



ICLR

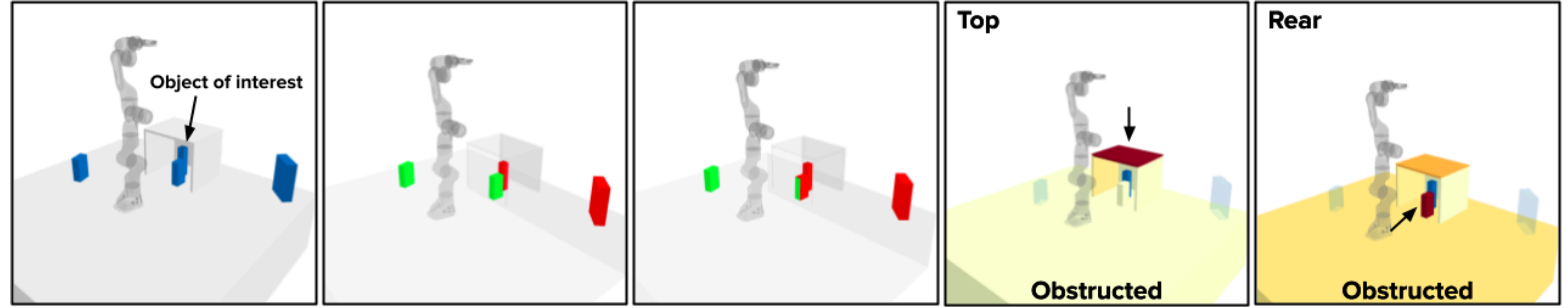
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Motivation

- Due to the **high combinatorial complexity** of **TAMP**, geometric planning creates a **bottleneck** to the planning process.
- Action feasibility prediction** offers an **efficient alternative** to geometric planning during the TAMP process.
- Existing approaches** to action feasibility prediction often **struggle with interpretability, scalability, and generalization** across diverse environment.
- Novel GNN-based model** for efficient and accurate action and grasp feasibility prediction in complex 3D environment.
- Mechanisms that improve feasibility prediction while **enhancing interpretability**

Problem Description

- Action and Grasp Feasibility Prediction:** Can the robot move the object? How?
 $F_a(\mathcal{O}, E) \in \mathbb{R}$ $F_g = [F_g(\mathcal{O}, E), \forall g \in \mathcal{G}] \in \mathbb{R}^5$
- Inverse Kinematics (IK) Feasibility Prediction:** Can the robot reach the object from each side?
 $\kappa_g = [\kappa_g(\mathcal{O}, E), \forall g \in \mathcal{G}] \in \mathbb{R}^5$
- Grasp Obstruction (GO) Estimation:** Ratio of grasps obstructed by neighboring objects for each side
 $\rho_g = [\rho_g(\mathcal{O}, \mathcal{O}'), \forall g \in \mathcal{G}, \forall \mathcal{O}' \in \mathcal{N}(\mathcal{O})] \in \mathbb{R}^{5 \times |\mathcal{N}(\mathcal{O})|}$



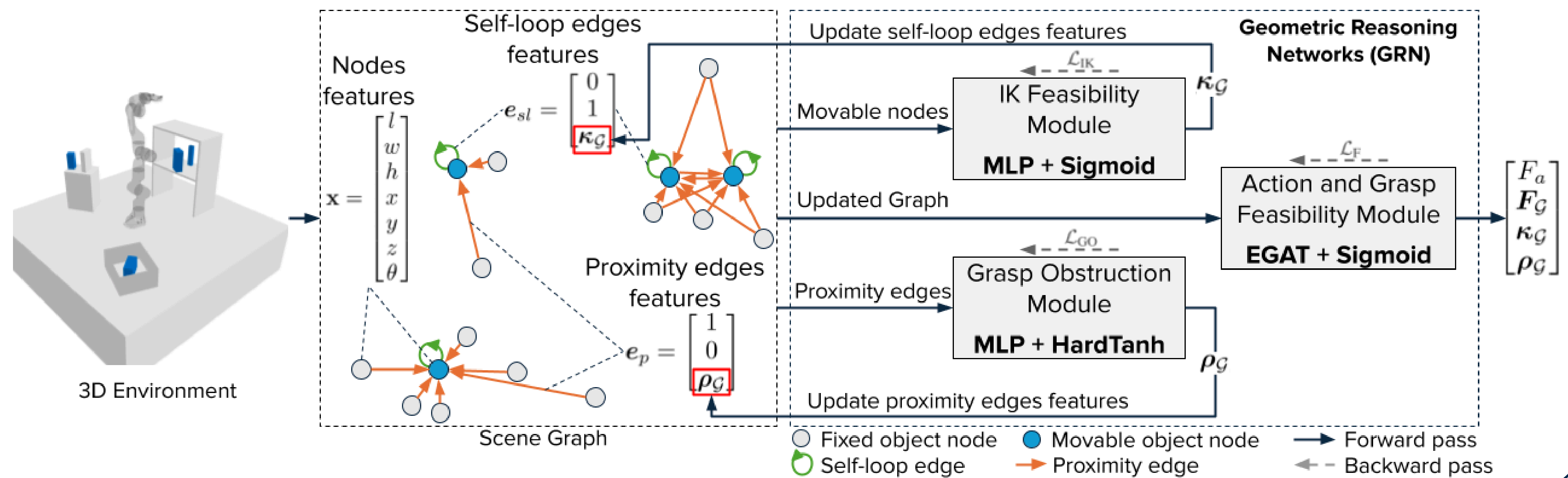
Geometric Reasoning Networks (GRN)

Graph-based scene representation:

- Nodes represent fixed and movable objects
- Two types of edges:
 - Proximity edges (Distance-based)
 - Self-loop edges

GRN is composed of 3 modules:

- IK feasibility predictions** used to update self-loop edges
- GO estimations** used to update proximity edges features
- An **EGAT** aggregates node and edge features to **predict action and grasp feasibility**

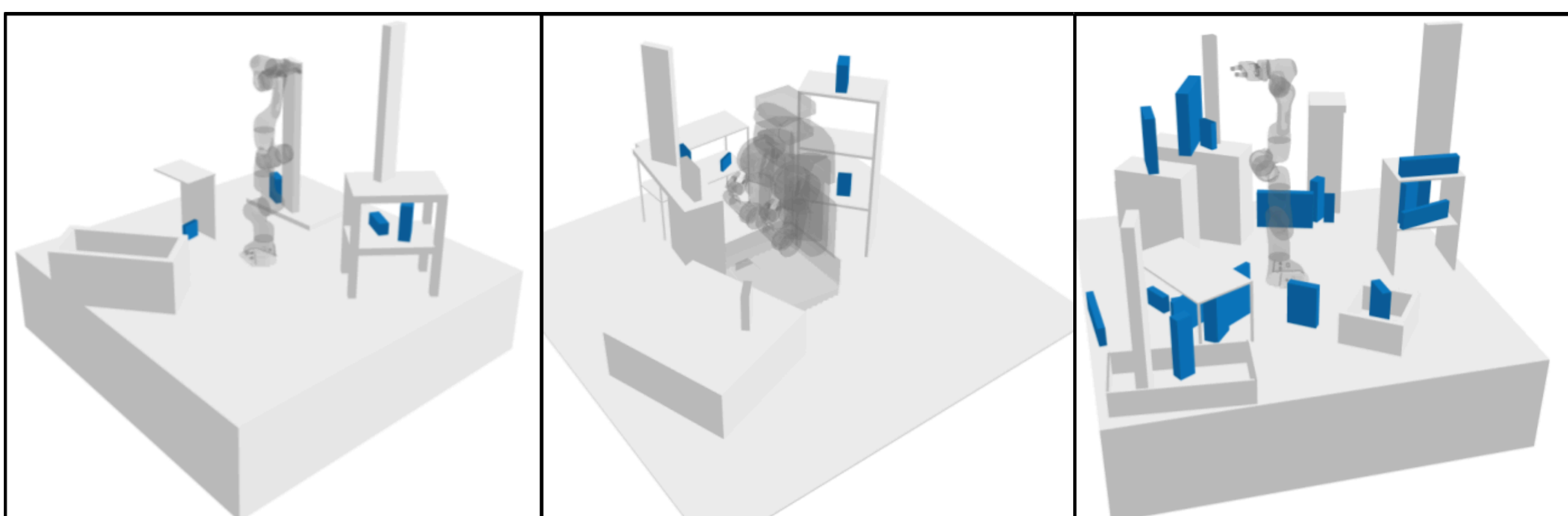


Training Strategy

- Each **GRN** module is pre-trained separately
- The complete **GRN** model is then fine-tuned using the following loss:

$$\mathcal{L} = \mathcal{L}_F + \mathcal{L}_{IK} + \eta \cdot \mathcal{L}_{GO}$$

- The model is trained on fully synthetic datasets consisting of environments containing 4 movable objects



Results

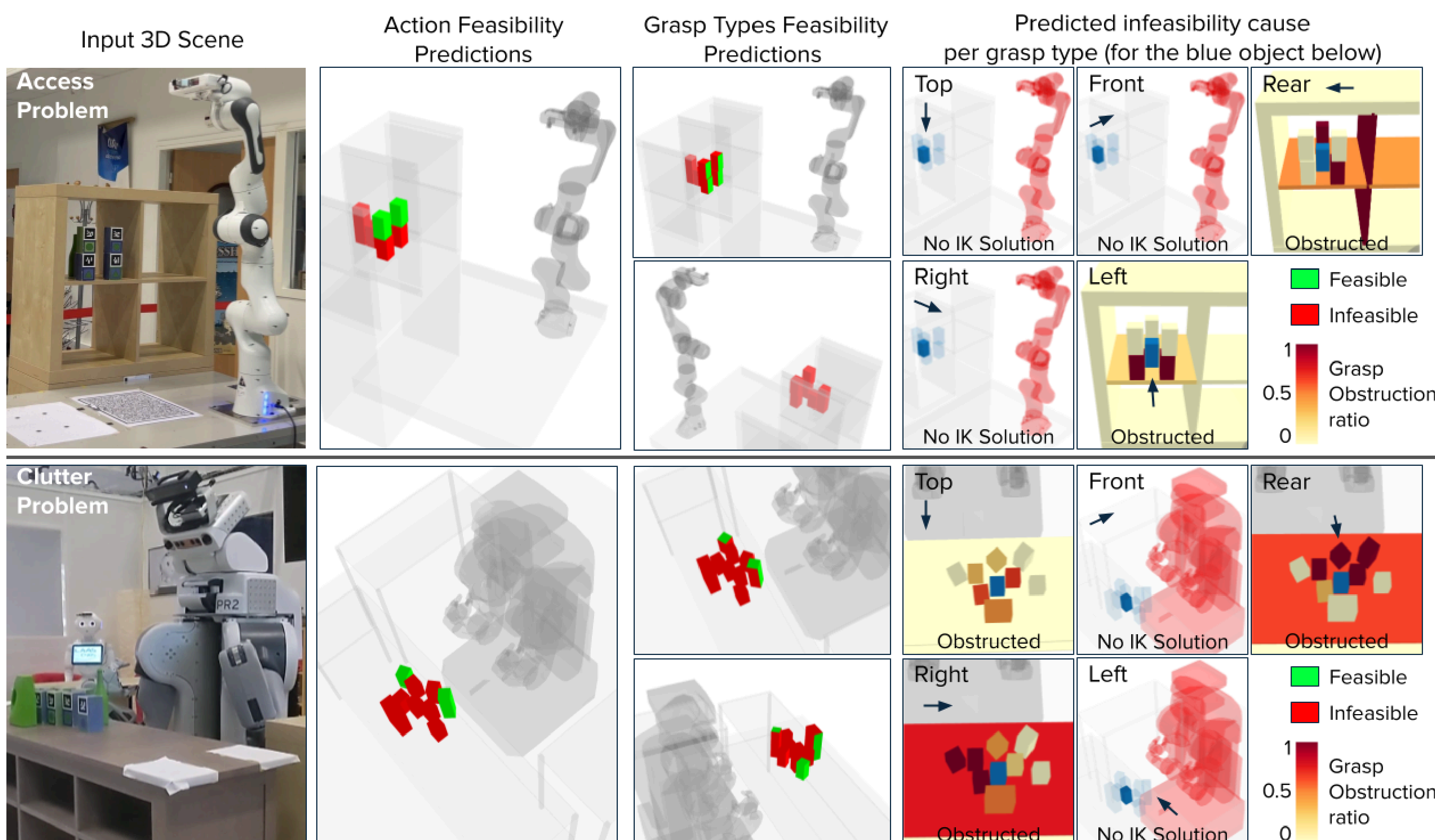
- GRN** achieve **state-of-the-art performance** in both action and grasp feasibility prediction.
- The model maintain accurate predictions for **robots with more complex kinematics** such as the PR2 robot.
- GRN** is able to generalize to environments with a **much higher number of objects**.

Dataset	Panda-3D-4		Panda-Tabletop-4		PR2-3D-4	
Task	Action (F1)	Grasp (F1)	Action (F1)	Grasp (F1)	Action (F1)	Grasp (F1)
DVH (Driess et al. 2020b)	0.840	0.718	0.961	0.865	0.808	0.622
F-GCN (Silver et al. 2021a)	0.836	0.721	0.955	0.879	0.791	0.680
F-GAT (Khodeir et al. 2023a)	0.867	0.796	0.961	0.904	0.827	0.764
AGFPNet (Bouhsain et al. 2023)	0.882	0.806	0.964	0.916	0.836	0.655
GRN (Ours)	0.939	0.940	0.976	0.976	0.908	0.903

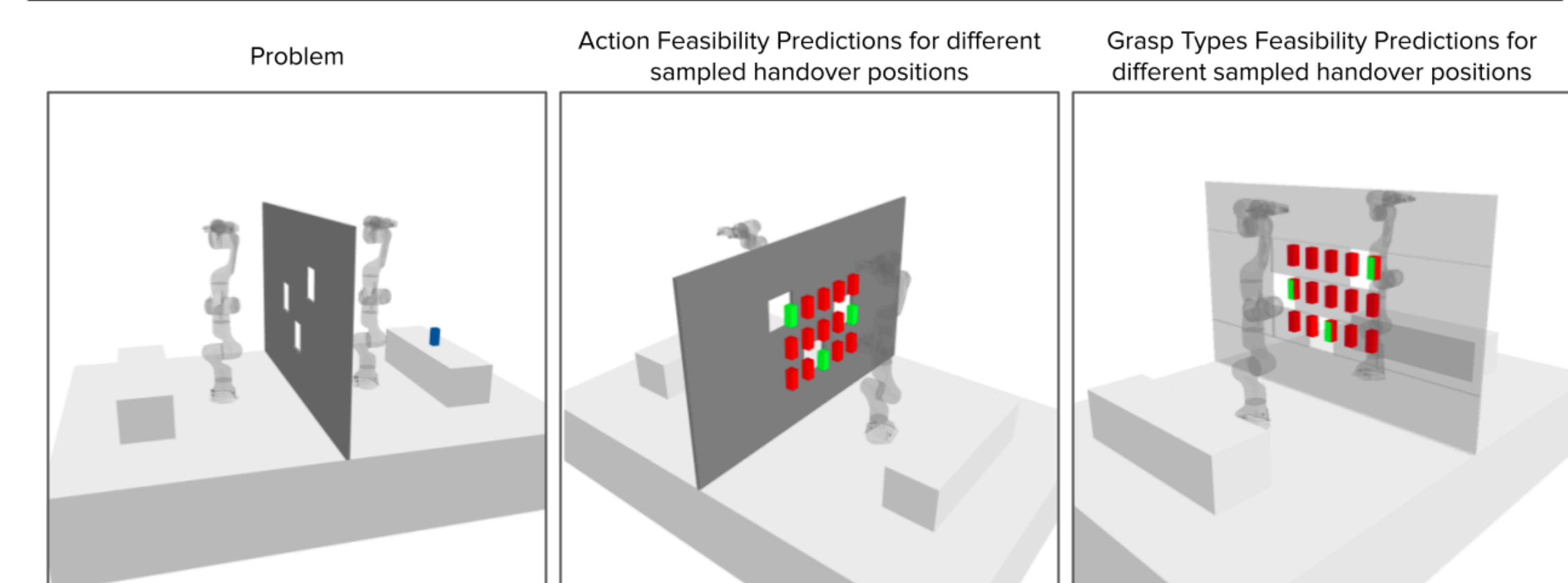
Test set	Panda-3D-10		Panda-3D-15		Panda-3D-20	
Task	Action (F1)	Grasp (F1)	Action (F1)	Grasp (F1)	Action (F1)	Grasp (F1)
DVH (Driess et al. 2020b)	0.820	0.697	0.819	0.686	0.825	0.676
AGFPNet (Bouhsain et al. 2023)	0.858	0.770	0.862	0.768	0.864	0.755
F-GCN (Silver et al. 2021a)	0.794	0.633	0.771	0.595	0.764	0.565
F-GAT (Khodeir et al. 2023a)	0.829	0.738	0.826	0.725	0.825	0.715
GRN (Ours)	0.894	0.909	0.891	0.906	0.890	0.903

Application to Task and Motion Planning

- Single query to GRN** → **geometrically feasible solutions** to complex TAMP problems by **tracing back the predicted reasons of infeasibility** for each object.
- Significantly faster planning time** compared to traditional TAMP methods, on **single and multi-robot problems**



Problem	Method	Success Rate	Planning time (s)	Nb Geometric Planner Calls
Access-monotonic	Baseline	100%	26.5	41.1
	GRN planner	90%	3.17	6
Clutter	Baseline	100%	558.9	89.5
	GRN planner	100%	14.8	7.2



References

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- Bouhsain et al. "Simultaneous Action and Grasp Feasibility Prediction for Task and Motion Planning through Multi-Task Learning." IROS 2023
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